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Introduction to Training Detection Models

Preparing Your Data for Training

Performing Transfer Learning for Object Detection

🎥

Video: Performing Transfer Learning with Object Detection Models6 min

📖

Reading: Transfer Learning for Fasteners Detection1h

📝

Quiz: Week 2 QuizSubmitted

Week 2 Quiz

Review Learning Objectives

✔ Submit your assignment

Due Feb 25, 11:59 PM IST

✔ Receive grade
To Pass 75% or higher

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1. How would you modify the size of images used to train an object detection model? 1 / 1 point

- ☐ Object detection data cannot be resized because each image is matched with labeled bounding boxes which have coordinates relative to the original images.
- ☒ Use the transform function along with a custom data resizing function to resize the images as well as scale and shift the bounding boxes associated with each image.
- ☐ Use an augmented image datastore to resize the images in your ground truth data to a new target size. Then proceed with training.
- ☒ Correct
We provide a `resizeDetectionData` function with the course files that you can use with the transform function to resize images and bounding boxes.

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100%
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2. Which statements below that are true about anchor boxes? Select **all** that apply. 1 / 1 point

- ☒ Anchor boxes are often estimated using a clustering to find a set of sizes and aspect ratios that best represent the ground truth bounding boxes.
- ☒ Correct
In MATLAB, this is done using the `estimateAnchorBoxes` function.
- ☒ Models that require anchor boxes use them both during training and when detecting objects after training.
- ☒ Correct
Anchor boxes are used while training a model and provide initial guesses when performing detection on images.
- ☐ Anchor boxes are always necessary to train an object detection model.
- ☐ Increasing the number of anchor boxes results in faster training times and prediction speeds.

3. Assume you are training a YOLO v4 model which requires anchor boxes. Which workflow below has the steps in the correct order? 1 / 1 point

(Note that not all steps are included for simplicity.)

- ☒ 1. Analyze the data
2. Resize the data
3. Estimate anchor boxes
4. Train a model
5. Evaluate the results
- ☐ 1. Analyze the data
2. Estimate anchor boxes
3. Resize the data
4. Train a model
5. Evaluate the results
- ☒ Correct
It is important to resize the data before estimating anchor boxes so that the anchor boxes align with data you use for training and inference.

4. What is the best approach to creating your ground truth for your image data in MATLAB? 1 / 1 point

- ☒ Use the Image Labeler App
- ☐ Manually find the pixel coordinates of bounding boxes for each object in your image and store them in a large array.
- ☐ Search the web for datasets that someone else has already labeled and use those images instead.
- ☒ Correct
This provide an interface for creating labels and bounding boxes. It also include tools to help you automate the process which you'll learn in course 3 of this specialization.

5. Assume you have a ground truth saved in the file trainingGT.mat 1 / 1 point

Which option has the correct code to load the ground truth and start preparing it for training an object detection model in MATLAB?

- ☐

```
1 [inds,blds] = loadObjectDetectorGroundTruth("trainingGT.mat");
2 trainingData = combine(inds,blds);
```
- ☐

```
1 trainingGT = importGroundTruth("trainingGT.mat");
2 [inds,blds] = objectDetectorTrainingData(trainingGT);
3 trainingData = combine(inds,blds);
```
- ☐

```
1 trainingData = importObjectDetectorTrainingData("trainingGT.mat");
```
- ☒

```
1 load trainingGT.mat trainingGT
2 [inds,blds] = objectDetectorTrainingData(trainingGT);
3 trainingData = combine(inds,blds);
```

✔ Correct

6. Which of the following options best describes the usefulness of performing analysis on your ground truth? 1 / 1 point

- ☐ It can help you determine which detection model frameworks may work best based on the distribution of object sizes.
- ☐ It can help you find potentially mislabeled data or other data issues by allowing you to spot outliers in area, aspect ratio, or other quantities you choose to analyze.
- ☐ It can help you determine if you have an imbalance of object instances by class, area, aspect ratio, or other quantities you choose to analyze.
- ☒ It can help you with all of the above.
- ☒ Correct

7. In the live script reading, AnalyzingYourLabeledData.mlx, you were asked to create a box plot to visualize distribution of Area for each class of fastener. What is the area for the largest Area outlier of any class? 1 / 1 point

- ☐ 7897
- ☐ 42
- ☐ 12.08
- ☐ 15738
- ☒ 12642
- ☒ Correct

8. Assume that the object classes, a set of anchor boxes, training options, and the required image size have already been defined using the variables. 1 / 1 point

- classes
- trainingAnchors
- options
- targetDataSize

Which two lines of code are necessary to train a YOLOv4 object detector using a pretrained network with the Darknet 53 backbone and a desired input size of `targetDataSize`?

- ☒

```
1 detector = yolov4ObjectDetector("csp-darknet53-coco",classes,trainingAnchors,"InputSize",targetDataSize);
2 [detector,info] = trainYOLOv4ObjectDetector(trainingData,detector,options);
```
- ☐

```
1 myYOLOv4Detector = yolov4ObjectDetector("csp-darknet53-coco",classes,trainingAnchors,"InputSize",targetDataSize);
2 [detector,info] = trainYOLOv4ObjectDetector(trainingData,"detector",myYOLOv4Detector,options);
```
- ☐

```
1 detectorType = yolov4(classes,trainingAnchors);
2 [detector,info] = trainYOLOv4ObjectDetector(trainingData,detectorType,"InputSize",targetDataSize);
```

✔ Correct
These two lines of code create the initial detector, and then retrain it using your training data.

